(For Study Purposes Only)

COMP 3721 – Practice Midterm Exam

Complete the following boxes in ink and ensure to write legibly. Any additional pages used for this exam need to be handed in inside this booklet. Read all rules, instructions, and questions carefully.

Candidates Name:

_____ *(Set):* _____

(First name) *(Surname)*

Signature: _____

Student ID Number:

Examination Protocols

1. All examinations will start promptly at the scheduled time of the examination.
2. No late arriving student will be admitted writing the test/exam after any student has left the exam room.
3. Only materials designated by the professor in the examination instructions are to remain on the desk.
4. All personal belongings, i.e., coats, book bags, pocketbooks, cell phones, tablets, etc. must be placed in a location at the front of the classroom.
5. No outerwear items are permitted to be worn (e.g. hats, gloves, overcoats, jackets, vests). Religious head coverings are exempted.
6. There is to be no food or drink in the classrooms, either before or during the exams. Only clear water in clear containers will be allowed in the classrooms.
7. Students will not speak with each other once they have entered the test/exam room.
8. Students are not permitted to share personal items (pens, calculators, etc.) during an examination.
9. Students may speak only to the professor or proctor during the test/exam
10. No more than one student shall leave the examination room at any time.
11. All students must stop typing or writing and submit their exam when the allotted time has expired
12. **The exam is 100 minutes.**
13. **Total Marks: 60 marks.**

PART A – T/F QUESTIONS (1 MARK EACH) [15]

Circle <u>one</u> answer clearly.

T F **A1.** The data-link layer in the TCP/IP model is responsible for routing packets between networks.

T F **A2.** A composite signal can be decomposed into multiple simple sine waves using Fourier analysis.

T F **A3.** Phase describes the position of a waveform relative to time zero.

T F **A4.** In baseband transmission, the signal occupies a band of frequencies around a carrier.

T F **A5.** The Shannon capacity formula accounts for noise in the channel.

T F **A6.** NRZ-L encoding uses a transition at the beginning of each bit interval to represent bit value 1.

T F **A7.** In 16-QAM, four bits are transmitted per signal element.

T F **A8.** In TDM, interleaving can be done at the bit level or at the byte level.

T F **A9.** Statistical TDM assigns time slots dynamically based on demand.

T F **A10.** The Nyquist formula gives the maximum theoretical data rate for a noisy channel.

T F **A11.** In NRZ-I encoding, long runs of zeros can cause synchronization loss.

T F **A12.** A bandpass channel has a bandwidth that does not start at zero.

T F **A13.** In FSK, the amplitude of the carrier signal changes to represent different bits.

T F **A14.** During encapsulation, each layer adds its own header (and sometimes trailer) to the data from the upper layer.

T F **A15.** In PSK, the number of phase shifts determines the number of bits per signal element.

4. Nyquist formula → Noiseless channel

4. Shannon formula → Noisy channel

Encoding Sync Problem?
NRZ-L Yes
NRZ-I Yes (long 0s)
Manchester No (always mid-bit transition)

Midterm Trap:

Synchronous TDM → fixed slots

Statistical TDM → dynamic slots

⚠ Exam Trap:
Modulation What changes?
ASK Amplitude
FSK Frequency
PSK Phase

⚠ Important:

Encapsulation happens at:

Sender side

PART B – MC QUESTIONS [2 MARKS EACH] [20]

Circle one answer clearly.

B1. Which layer of the TCP/IP model is responsible for host-to-host delivery?

A) Application

B) Transport

C) Network

D) Data Link

B2. A sine wave completes 500 cycles in one second. Its period is:

A) 500 s

B) 2 ms

C) 0.5 ms

D) 0.002 s

B3. A signal is amplified by a factor of 1000 in power. The gain in dB is:

A) 10 dB

B) 20 dB

C) 30 dB

D) 100 dB

B4. If a modulation scheme uses 64 signal levels, the number of bits per signal element is:

A) 4

B) 5

C) 6

D) 8

B5. The Nyquist sampling theorem states that the minimum sampling rate must be:

A) Equal to the maximum frequency

B) Equal to the bandwidth

C) Twice the maximum frequency

D) Four times the maximum frequency

B6. Which digital-to-analog conversion technique varies the frequency of the carrier?

A) ASK

B) FSK

C) PSK

D) QAM

B7. In synchronous TDM, empty slots occur when:

A) The channel is noisy

B) A source has no data to send

C) Guard bands overlap

D) Bandwidth is insufficient

B8. The main purpose of ARP is to:

A) Route packets across networks

B) Resolve domain names to IP addresses

C) Map IP addresses to MAC addresses

D) Encrypt data-link frames

B9. Which encoding scheme guarantees a transition at the middle of every bit interval?

A) NRZ-L

B) NRZ-I

C) Manchester

D) Bipolar AMI

B10. A signal has peak voltage V1 = 20V at the sender and V2 = 2V at the receiver. The attenuation in dB is:

A) -10 dB

B) -20 dB

C) -6 dB

D) -40 dB

PART C – SHORT QUESTIONS [10]

C1 (2 marks): Explain the difference between baseband and broadband transmission.

Baseband transmission sends digital signals directly over a low-pass channel whose bandwidth starts at 0 Hz. It does not use a carrier signal.

Broadband transmission uses modulation to send signals over a bandpass channel centered around a carrier frequency.

C2. (2 marks): List the five components of a data communication system.

Message – the information to be communicated

Sender – the device that sends the message

Receiver – the device that receives the message

Transmission medium – the physical path

Protocol – the rules governing communication

C3. (2 marks): Explain the difference between synchronous TDM and statistical TDM.

In synchronous TDM, each input source is assigned a fixed time slot in every frame, even if it has no data to send, which may result in empty slots.

In statistical TDM, time slots are assigned dynamically based on demand, so only active sources are given slots, improving efficiency.

C4. (2 mark): Explain self-synchronization in digital encoding. Give one example of an encoding scheme that provides it.

Self-synchronization means the receiver can recover the sender's clock from the received signal without needing a separate timing signal. This is achieved by ensuring re

Example: Manchester encoding guarantees a transition in the middle of every bit interval, providing self-synchronization.

C5. (2 marks): Explain the role of encapsulation and decapsulation in the TCP/IP model.

Encapsulation is the process in which each layer of the TCP/IP model adds its own header (and sometimes trailer) to the data received from the upper layer before transmission.

Decapsulation is the reverse process at the receiver, where each layer removes its corresponding header to retrieve the original data.

PART D – ESSAY QUESTIONS [15]

D1 (3 marks): A signal travels over a channel where the bandwidth starts at 500 kHz and ends at 1 MHz. Determine the type of this channel. Explain why digital signals cannot be sent directly over this channel and state what type of conversion is needed.

The channel is a bandpass channel because its bandwidth does not start at 0 Hz

Digital signals cannot be sent directly since they require a low-pass channel.

Therefore, digital-to-analog conversion (modulation) is required before transmission.

D2 (2 marks): The frequency-domain representation of a composite signal shows discrete vertical lines at frequencies 1 kHz, 3 kHz, and 5 kHz. Determine whether this signal is periodic or non-periodic. Justify your answer.

The signal is periodic because discrete vertical lines in the frequency domain indicate distinct frequency components, which occur only in periodic signals.

D3. (2 marks): Four voice channels each require 3 kHz bandwidth. They are multiplexed using FDM with 500 Hz guard bands between adjacent channels. Find the minimum required total bandwidth and show the frequency allocation.

Number of guard bands= =4 − 1=3.

Btotal=4(3 kHz)+3(0.5 kHz)=12 kHz+1.5 kHz=13.5 kHz

Minimum required total bandwidth = 13.5kHz

Frequency allocation example (start at 0 kHz):
Ch1: 0–3 kHz, GB: 3–3.5 kHz
Ch2: 3.5–6.5 kHz, GB: 6.5–7 kHz
Ch3: 7–10 kHz, GB: 10–10.5
Ch4: 10.5–13.5 kHz

D4 (4 marks): A channel has bandwidth = 4 MHz and SNR = 63.

a. Compute the Shannon capacity.

b. What is the SNR in dB?

c. If we use 32-QAM, what is r (bits per signal element)?

d. Compute the Nyquist maximum data rate for this L. Compare with Shannon capacity.

Blog2(1+SNR).

D5 (4 marks):

We are using synchronous TDM to multiplex two channels. Each source has a data rate of 1 Mbps. If the input unit is 1 bit (i.e., each time slot is 1 bit) find the following:

a. The frame rate.

b. The duration of each frame.

c. The number of bits in each frame.

d. The data rate of the link.

Each frame has 1 bit from each source → 2 bits per frame.

Each source must send 1,000,000 bits/sec, and contributes 1 bit per frame.

So frame rate must be:
Frame rate = 1,000,000 frames/sec

Tframe=1/frame rate=1/1,000,000=10^−6 s= 1 microsecond (1 μs)

c. 2 sources × 1 bit each = 2 bits per frame

d. R=2×1,000,000=2,000,000 bps

— End of the Midterm Exam —

[EXTRA PAGE]

Given: Bandwidth=4 MHz
B=4 MHz, SNR = 63.

C = B log2(1 + SNR)

C = 4 × 10^6 × log2(1 + 63)
C = 4 × 10^6 × log2(64)
C = 4 × 10^6 × 6
C = 24 × 10^6 bps
Shannon Capacity = 24 Mbps

(b) SNR in dB
Formula:
SNR(dB) = 10 log10(SNR)
SNR(dB) = 10 log10(63)
SNR(dB) ≈ 10 × 1.799
SNR(dB) ≈ 18 dB

(c) r for 32-QAM
Formula:
r = log2(L)

r = log2(32)
r = 5 bits per signal element

(d) Nyquist Maximum Data Rate

Formula:
BitRate = 2B log2(L)

BitRate = 2 × 4 × 10^6 × 5
BitRate = 40 × 10^6 bps

Nyquist maximum data rate = 40 Mbps

Comparison:
Shannon capacity = 24 Mbps
Nyquist = 40 Mbps

Since the channel is noisy, the actual maximum data rate is limited by Shannon.
Therefore, the practical maximum rate is 24 Mbps.

Formula Sheet

- $C(n, b) = \frac{n!}{b!(n-b)!}$
- $s(t) = A\sin(2\pi ft) = A\sin(2\frac{\pi}{T}t) = A\sin(\omega t + \varphi)$
- $\omega = 2\pi f$
- Period: $T = 1/f$
- $\lambda = c/f$
- $360^o = 2\pi$ (rad)
- Bit length = propagation speed × bit duration
- Bit duration = 1/Bit rate
- $B\ (Hz) = f_h - f_l$
- Attenuation (Amplification): $dB = 10 \log_{10} \frac{P2}{P1} = 20 \log_{10} \frac{V_2}{V_1}$
- $SNR = \frac{\text{Average Signal Power}}{\text{Average Noise Power}}$
- $SNRdb = 10 \log_{10} SNR$
- $BitRate(bps) = 2 \cdot bandwidth\ (B) \cdot \log_2 L$
- $Capacity(bps) = bandwidth\ (B) \cdot \log_2 (1 + SNR)$
- Number of bits per levels: $\log_2 L$
- $S = c \cdot N \cdot 1/r$
- $B = c \cdot N \cdot 1/r$
- $S = N/r$
- Nyquist rate: $f_n = 2 \cdot f_m$
- Nyquist interval: $T_n = 1/f_n = 1/2f_m$
- PCM Bandwidth: $B_{min} = n_b \cdot B_{analog}$
- $\Delta = \frac{V\max - V\min}{L}$
- Bandwidth for ASK and PSK: $B = (1 + d) \cdot S$
- Bandwidth for FSK: $B = (1 + d) \cdot S + 2\Delta f$